

Detecting Partisan Gerrymandering through Ensemble Analysis of Mid-Decade Redistricting

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About the Learner

The first time my Texas model finished a full ensemble, the result looked almost too clean: the enacted 2025 map produced eight Democratic seats, while none of 500 population-balanced, contiguous alternatives produced fewer than ten. I had seen a result look this perfect before and had learned to be careful of it. Studying abroad, my predictions of whether European labor-market sectors would grow or shrink came out accurate to within a percent on an 80/20 train-test split, and my professor flagged it. So, my first reaction to Texas was not to treat the result as proof, but to ask what my assumptions could be making it look that way.

I arrived at that question by trying to connect politics with a part of data science I had never used before. I started with 3Blue1Brown and StatQuest videos on Markov chains and Monte Carlo methods, then followed the trail into the redistricting literature, especially the ReCom method developed by DeFord, Duchin, and Solomon. I then built a working proof of concept in Python with GerryChain, GeoPandas, pandas, census data, and precinct-level election results. The pipeline generates alternative congressional maps and compares their partisan outcomes enacted plan.¹²

At this point, I am confident in cleaning geographic and election data, building the pipeline, running ReCom chains, and recognizing when a clean result brings skepticism. I am less confident in what comes next: deciding whether a chain explored the map space well enough, which constraints belong in the ensemble, and how much a conclusion depends on one election or modeling choice. Those are not details I want to leave unanswered; they're vital for the validity of this project. They are the research questions I want this fellowship to help me answer.

Dr. Eric Gerber and I will establish regular meetings at the start of the fellowship, with additional check-ins before major modeling decisions. I will bring intermediate results and specific questions, especially when choosing constraints, interpreting sensitivity tests, and deciding what the evidence can reasonably support. Three learning objectives will emerge from this project:

- 1. Understand ensemble sampling beyond calling a library.** I will study MCMC diagnostics, run independent chains from different starting maps, and compare their behavior before treating an ensemble as informative.
- 2. Learn what each redistricting rule changes.** I will add constraints such as compactness and political-boundary splits one at a time, document each choice, and measure how each shifts the distribution rather than assuming stricter rules to add to the detection of my algorithm.
- 3. Communicate uncertainty without hiding the result.** I will test explanations with non-specialists and design the public tool, so assumptions and limitations are beside the finding, not several clicks away from it, and not difficult for non-specialists in this field to digest.

Purpose and Significance

The project asks this question: when a state adopts a congressional map, is its partisan outcome something neutral redistricting processes regularly produce, or is it an outlier? I generate many alternative maps under a defined set of neutral rules, measure the partisan result of each, and place the enacted map inside that distribution. An extreme result does not prove intent. It shows that, under the rules being tested, the enacted outcome is unusually difficult to reproduce.

Texas is the proof of concept, not the endpoint. The broader value is a transparent baseline for a debate that often leans on maps that look suspicious or partisan outcomes presented alone. Ensemble analysis asks a harder question: compared with what? A public tool can make that comparison inspectable by people who do not work in computational redistricting, while keeping the assumptions visible enough for them to question the result.

Objectives and Methods

First, I will strengthen the Texas model. I will audit the pipeline from raw data through district generation and outcome calculation, then test how data-cleaning choices affect the result. My precinct-level census and election

¹Daryl DeFord, Moon Duchin, and Justin Solomon, "Recombination: A Family of Markov Chains for Redistricting," *Harvard Data Science Review* 3, no. 1 (2021).

²GerryChain documentation, "Running a chain with ReCom," gerrychain.readthedocs.io.

data come from the Redistricting Data Hub. Because precinct boundaries shifted between the 2020 census and 2024 election, I used spatial joins to combine population and voting data. Roughly 1,500 precinct units currently receive zero population, likely because they represent water or unpopulated areas. I will verify those cases rather than assume they are harmless.³

Next, I will expand the ensemble one assumption at a time. The current model enforces roughly equal population and contiguity. I will test compactness and county or municipal splits where the data permit it, then compare how each constraint changes the distribution. I will not assume that stricter rules automatically make an outlier more extreme; they can move the whole distribution instead. I will also test multiple statewide elections where comparable precinct data exist, rather than mindlessly letting one presidential election stand in for every expression of partisanship.

Second, I will make the method portable. After validating Texas, I will adapt the pipeline to other states that were redistricted during this cycle. Each state brings different data, district counts, legal criteria, and geographic preprocessing, so I will build a reproducible state configuration rather than copy Texas blindly. For each state, I will run several thousand maps after validation, compare independent chains, and document instability before publishing a result.

Third, I will build a public research tool. The dashboard will place each enacted outcome against its simulated distribution, explain which rules are generated by the ensemble, and show when conclusions change assumptions. I will use open-source Python tools, Redistricting Data Hub data, Northeastern computing resources if larger runs exceed my local capacity, and Dr. Gerber's guidance for this.

The project will proceed in four stages: weeks 1–4 validating Texas and testing assumptions; weeks 5–12 adapting the pipeline to additional states; weeks 13–18 running robustness checks, cross-state comparisons, and mentor review; and weeks 19–22 building the dashboard, documentation, and presentation materials. The biggest risks are computational cost, inconsistent state data, and a chain that looks productive without sampling the space well. If a state lacks reliable data or a model behaves poorly, I will narrow the scope rather than publish a result I cannot defend.

Outcomes, Evaluation, and Dissemination

I will hold the project to explicit checkpoints so I can adjust (or narrow) the scope. Texas is a validation state: I will not move to a new state until the Texas model reproduces the enacted seat outcome, and independent chains agree on whether the map is an outlier. For each additional state, I will not report a result until its ensembles are stable; independent chains agree, and the conclusion holds under more than one statewide election. A state that cannot clear those checks gets set aside, not published.

The primary outcome will be a public research tool showing where enacted congressional maps fall within ensembles of neutral alternatives. Supporting outcomes will include a reproducible codebase, documented state analyses, and a written methodology that explains both what the model measures and what it does not.

I will evaluate the project on three levels. **Technically**, results should be reproducible and reasonably consistent across independent chains. **Substantively**, I will test whether conclusions survive, weaken, or reverse when I change constraints and election inputs. **Publicly**, a non-specialist should understand what an outlier means without coming away with a stronger claim than the model supports. If an assumption changes the conclusion, that is not a failed result; it is information the tool should show.

I will present the project at RISE and publish the dashboard, code, and documentation through GitHub Pages. With Dr. Gerber, I will also assess whether the final analysis is appropriate for an external conference, student research venue, or publication. I do not want to build a tool that labels maps good or bad. The point is to make the argument testable, the assumptions visible, and the uncertainty just as easy to see as the headline result.

³Redistricting Data Hub, Texas data and precinct-boundary/election-result resources, redistrictingdatahub.org.